Capstone: Agentic IT Operations Platform

AI Solution Architect Program | Friday 16 October 2026 afternoon | 90-minute case study, then presentations

The case

A global enterprise wants an Agentic AI-powered IT Operations Platform that can autonomously manage production incidents. The system should: monitor alerts from multiple systems; analyse logs and telemetry; retrieve troubleshooting SOPs; identify probable root causes; execute approved remediation workflows; maintain memory of past incidents; coordinate across multiple specialised AI agents; and interact with enterprise tools using MCP.

Example scenario

A payment application shows high latency, an increased error rate and database timeout exceptions. The platform should: 1 detect the anomaly; 2 gather logs and metrics; 3 retrieve similar historical incidents; 4 generate a root-cause hypothesis; 5 recommend remediation; 6 execute safe automation steps; 7 escalate if confidence is low.

Expected agents

Monitoring Agent, Log Analysis Agent, RAG Retrieval Agent, Decision Agent, Automation Agent, Compliance Agent, Escalation Agent.

Areas to focus on

LLM selection, prompt engineering, vector database, RAG, guardrails, agentic AI, design patterns, short-term memory, long-term memory, MCP. Mention your tech stack based on your knowledge.

Deliverables and scoring

Deliverable	Weight
1 End-to-end architecture diagram	50%
2 How you handle these scenarios: why multi-agent; which design pattern; memory strategy; human-in-the-loop boundaries; tool calling strategy; failure handling; cost optimization; observability of agents	30%
3 Description of each layer in the diagram and the concept used	20%

Suggested use of the 90 minutes

Time	Activity
10 min	Read the case; split roles; list the layers you need
50 min	Draw the architecture diagram (spend about half your time here)
30 min	Write the eight scenario answers and the layer descriptions (use the answer template)

Presentations

After the break, each team presents for about 8 minutes: the diagram, the key trade-offs, and your answers to the eight questions.

Ground rules

- Use plain technology names and say why you chose them (any framework or cloud is fine).
- Every choice needs a reason: say what you rejected and why.
- Use what you built this week: tools, MCP, memory, RAG, monitoring, guardrails.
- Use synthetic or public information only.